



Join us at javatechcommunity.com for our Premium Java Interview Kit and get referrals for Java-based roles – absolutely **FREE!**

DAY 7 /100

JAVA INTERVIEW - CODING ROUND



TOP-10

STRING PROGRAMS

- 01 Write a program to reverse a string.

- 02 Write a program to check if a string is a palindrome.

- 03 Write a program to count vowels and consonants in a string.

- 04 Write a program to find the first non-repeated character in a string.

- 04 Write a program to check if two strings are anagrams.

- 06 Write a program to remove all whitespaces from a string.

- 07 Write a program to count the occurrences of a character in a string.

- 08 Write a program to check if a string contains another string.

- 09 Write a program to find the maximum occurring character in a string.

- 10 Write a program to find the length of a string without using a function.



Join us at javatechcommunity.com for our Premium Java Interview Kit and get referrals for Java-based roles – absolutely **FREE!**

01. Write a program to reverse a string

With String Methods

javaCopy

```
public class ReverseStringWithMethods {
    public static void main(String[] args) {
        String str = "JavaTechCommunity";
        String reversed = new StringBuilder(str).reverse().toString();
        System.out.println("Reversed String: " + reversed);
    }
}
```

Without Built-in Methods

javaCopy

```
public class ReverseStringWithoutMethods {
    public static void main(String[] args) {
        String str = "JavaTechCommunity";
        char[] chars = str.toCharArray();
        String reversed = "";

        for (int i = chars.length - 1; i >= 0; i--) {
            reversed += chars[i];
        }
        System.out.println("Reversed String: " + reversed);
    }
}
```

02. Write a program to check if a string is a palindrome

With String Methods

javaCopy

```
public class PalindromeCheckWithMethods {
    public static void main(String[] args) {
        String str = "madam";
        String reversed = new StringBuilder(str).reverse().toString();

        if (str.equals(reversed)) {
            System.out.println(str + " is a palindrome");
        } else {
            System.out.println(str + " is not a palindrome");
        }
    }
}
```

Without Built-in Methods

javaCopy

```
public class PalindromeCheckWithoutMethods {
    public static void main(String[] args) {
        String str = "madam";
        boolean isPalindrome = true;
        int length = str.length();

        for (int i = 0; i < length / 2; i++) {
            if (str.charAt(i) != str.charAt(length - i - 1)) {
                isPalindrome = false;
                break;
            }
        }

        if (isPalindrome) {
            System.out.println(str + " is a palindrome");
        } else {
            System.out.println(str + " is not a palindrome");
        }
    }
}
```



Join us at javatechcommunity.com for our Premium Java Interview Kit and get referrals for Java-based roles – absolutely **FREE!**

03. Write a program to count vowels and consonants in a string

With String Methods

javaCopy

```
public class CountVowelsConsonantsWithMethods {
    public static void main(String[] args) {
        String str = "JavaTechCommunity";
        str = str.toLowerCase();
        int vowels = 0, consonants = 0;

        for (char ch : str.toCharArray()) {
            if ("aeiou".indexOf(ch) != -1) {
                vowels++;
            } else if (ch >= 'a' && ch <= 'z') {
                consonants++;
            }
        }
        System.out.println("Vowels: " + vowels);
        System.out.println("Consonants: " + consonants);
    }
}
```

Without Built-in Methods

javaCopy

```
public class CountVowelsConsonantsWithoutMethods {
    public static void main(String[] args) {
        String str = "JavaTechCommunity";
        str = str.toLowerCase();
        int vowels = 0, consonants = 0;

        for (int i = 0; i < str.length(); i++) {
            char ch = str.charAt(i);
            if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
                vowels++;
            } else if (ch >= 'a' && ch <= 'z') {
                consonants++;
            }
        }
        System.out.println("Vowels: " + vowels);
        System.out.println("Consonants: " + consonants);
    }
}
```



Join us at javatechcommunity.com for our Premium Java Interview Kit and get referrals for Java-based roles – absolutely **FREE!**

04. Write a program to find the first non-repeated character in a string

With String Methods

java

Copy

```
import java.util.LinkedHashMap;
import java.util.Map;

public class FirstNonRepeatedCharWithMethods {
    public static void main(String[] args) {
        String str = "swiss";
        Map<Character, Integer> charCountMap = new LinkedHashMap<>();

        for (char ch : str.toCharArray()) {
            charCountMap.put(ch, charCountMap.getOrDefault(ch, 0) + 1);
        }

        for (Map.Entry<Character, Integer> entry : charCountMap.entrySet()) {
            if (entry.getValue() == 1) {
                System.out.println("First non-repeated character: " + entry.getKey());
                break;
            }
        }
    }
}
```

Without Built-in Methods

java

Copy

```
public class FirstNonRepeatedCharWithoutMethods {
    public static void main(String[] args) {
        String str = "swiss";
        char[] chars = str.toCharArray();
        boolean isRepeated;

        for (int i = 0; i < chars.length; i++) {
            isRepeated = false;
            for (int j = 0; j < chars.length; j++) {
                if (i != j && chars[i] == chars[j]) {
                    isRepeated = true;
                    break;
                }
            }
            if (!isRepeated) {
                System.out.println("First non-repeated character: " + chars[i]);
                break;
            }
        }
    }
}
```




Join us at javatechcommunity.com for our Premium Java Interview Kit and get referrals for Java-based roles – absolutely **FREE!**

05. Write a program to check if two strings are anagrams

With String Methods

```
java Copy

import java.util.Arrays;

public class CheckAnagramWithMethods {
    public static void main(String[] args) {
        String str1 = "listen";
        String str2 = "silent";

        char[] arr1 = str1.toCharArray();
        char[] arr2 = str2.toCharArray();

        Arrays.sort(arr1);
        Arrays.sort(arr2);

        if (Arrays.equals(arr1, arr2)) {
            System.out.println(str1 + " and " + str2 + " are anagrams");
        } else {
            System.out.println(str1 + " and " + str2 + " are not anagrams");
        }
    }
}
```

Without Built-in Methods

```
java Copy

public class CheckAnagramWithoutMethods {
    public static void main(String[] args) {
        String str1 = "listen";
        String str2 = "silent";

        if (str1.length() != str2.length()) {
            System.out.println(str1 + " and " + str2 + " are not anagrams");
            return;
        }

        int[] charCount = new int[26];

        for (int i = 0; i < str1.length(); i++) {
            charCount[str1.charAt(i) - 'a']++;
            charCount[str2.charAt(i) - 'a']--;
        }

        for (int count : charCount) {
            if (count != 0) {
                System.out.println(str1 + " and " + str2 + " are not anagrams");
                return;
            }
        }
        System.out.println(str1 + " and " + str2 + " are anagrams");
    }
}
```



Join us at javatechcommunity.com for our Premium Java Interview Kit and get referrals for Java-based roles – absolutely **FREE!**

06. Write a program to remove all whitespaces from a string

With String Methods

javaCopy

```
public class RemoveWhitespacesWithMethods {
    public static void main(String[] args) {
        String str = "Java Tech Community";
        String result = str.replaceAll("\\s", "");
        System.out.println("String without whitespaces: " + result);
    }
}
```

Without Built-in Methods

javaCopy

```
public class RemoveWhitespacesWithoutMethods {
    public static void main(String[] args) {
        String str = "Java Tech Community";
        String result = "";

        for (int i = 0; i < str.length(); i++) {
            if (str.charAt(i) != ' ') {
                result += str.charAt(i);
            }
        }
        System.out.println("String without whitespaces: " + result);
    }
}
```

07. Write a program to count the occurrences of a character in a string

With String Methods

javaCopy

```
public class CountCharacterOccurrencesWithMethods {
    public static void main(String[] args) {
        String str = "JavaTechCommunity";
        char target = 'm';
        int count = str.length() - str.replace(String.valueOf(target), "").length();
        System.out.println("Occurrences of " + target + ": " + count);
    }
}
```

Without Built-in Methods

javaCopy

```
public class CountCharacterOccurrencesWithoutMethods {
    public static void main(String[] args) {
        String str = "JavaTechCommunity";
        char target = 'm';
        int count = 0;

        for (int i = 0; i < str.length(); i++) {
            if (str.charAt(i) == target) {
                count++;
            }
        }
        System.out.println("Occurrences of " + target + ": " + count);
    }
}
```



Join us at javatechcommunity.com for our Premium Java Interview Kit and get referrals for Java-based roles – absolutely **FREE!**

08. Write a program to check if a string contains another string

With String Methods

javaCopy

```
public class StringContainsWithMethods {
    public static void main(String[] args) {
        String str = "JavaTechCommunity";
        String subStr = "Tech";

        if (str.contains(subStr)) {
            System.out.println("'" + str + "' contains '" + subStr + "'");
        } else {
            System.out.println("'" + str + "' does not contain '" + subStr + "'");
        }
    }
}
```

Without Built-in Methods

javaCopy

```
public class StringContainsWithoutMethods {
    public static void main(String[] args) {
        String str = "JavaTechCommunity";
        String subStr = "Tech";
        boolean contains = false;

        for (int i = 0; i <= str.length() - subStr.length(); i++) {
            if (str.substring(i, i + subStr.length()).equals(subStr)) {
                contains = true;
                break;
            }
        }

        if (contains) {
            System.out.println("'" + str + "' contains '" + subStr + "'");
        } else {
            System.out.println("'" + str + "' does not contain '" + subStr + "'");
        }
    }
}
```

09. Write a program to find the length of a string without using a function

With String Methods

javaCopy

```
public class StringLengthWithMethods {
    public static void main(String[] args) {
        String str = "JavaTechCommunity";
        System.out.println("Length of the string: " + str.length());
    }
}
```

Without Built-in Methods

javaCopy

```
public class StringLengthWithoutMethods {
    public static void main(String[] args) {
        String str = "JavaTechCommunity";
        int length = 0;

        for (char ch : str.toCharArray()) {
            length++;
        }
        System.out.println("Length of the string: " + length);
    }
}
```




Join us at javatechcommunity.com for our Premium Java Interview Kit and get referrals for Java-based roles – absolutely **FREE!**

10. Write a program to find the maximum occurring character in a string

With String Methods

java

 Copy

```
import java.util.HashMap;
import java.util.Map;

public class MaxOccurringCharWithMethods {
    public static void main(String[] args) {
        String str = "JavaTechCommunity";
        Map<Character, Integer> charCountMap = new HashMap<>();
        char maxChar = ' ';
        int maxCount = 0;

        for (char ch : str.toCharArray()) {
            charCountMap.put(ch, charCountMap.getOrDefault(ch, 0) + 1);
        }

        for (Map.Entry<Character, Integer> entry : charCountMap.entrySet()) {
            if (entry.getValue() > maxCount) {
                maxChar = entry.getKey();
                maxCount = entry.getValue();
            }
        }
        System.out.println("Maximum occurring character: " + maxChar + " (" + maxCount + " times)");
    }
}
```

Without Built-in Methods

java

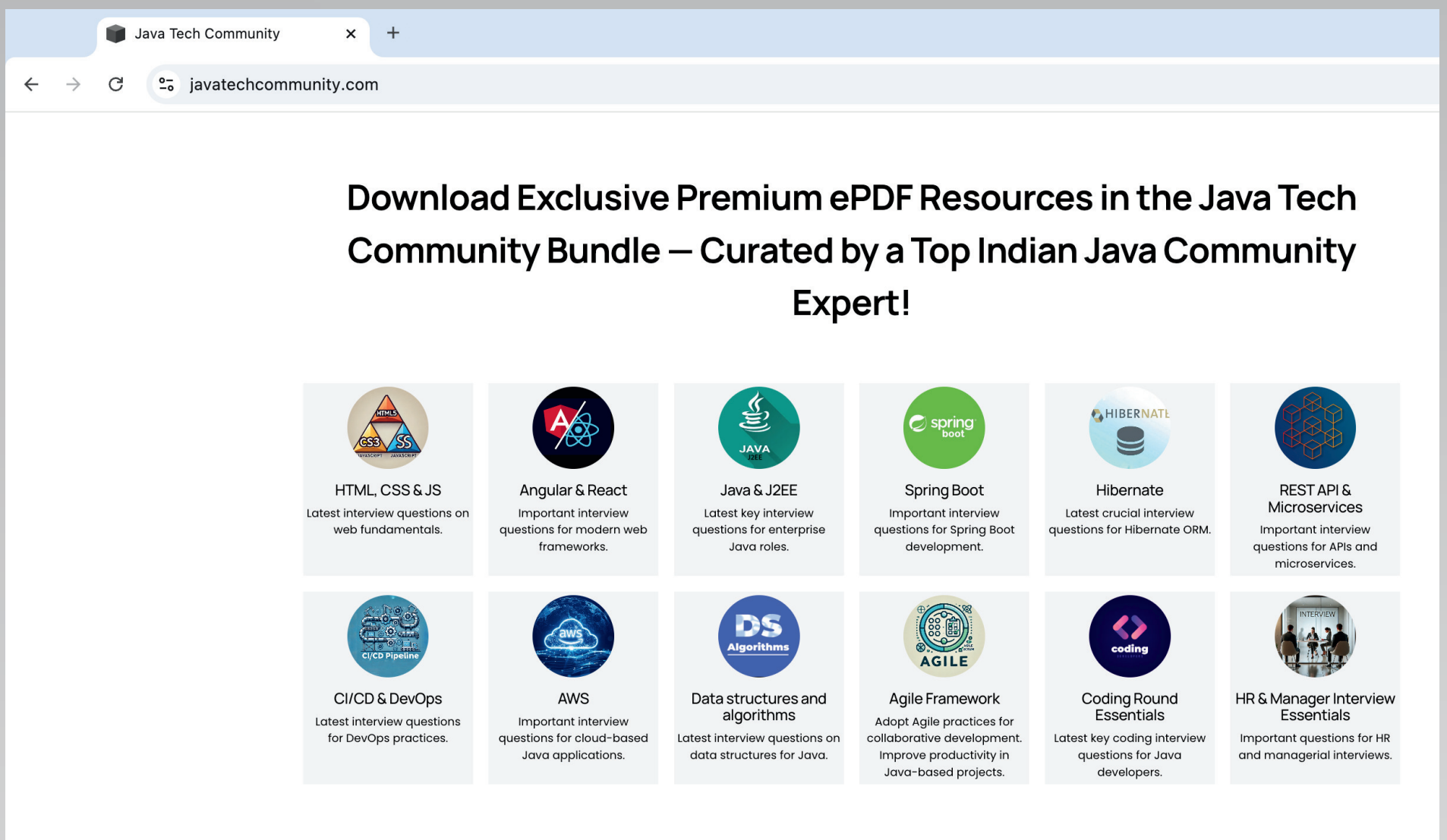
 Copy

```
public class MaxOccurringCharWithoutMethods {
    public static void main(String[] args) {
        String str = "JavaTechCommunity";
        int[] charCount = new int[256];
        char maxChar = ' ';
        int maxCount = 0;

        for (int i = 0; i < str.length(); i++) {
            charCount[str.charAt(i)]++;
        }

        for (int i = 0; i < charCount.length; i++) {
            if (charCount[i] > maxCount) {
                maxCount = charCount[i];
                maxChar = (char) i;
            }
        }
        System.out.println("Maximum occurring character: " + maxChar + " (" + maxCount + " times)");
    }
}
```


DO YOU WANT ACCESS TO ALL THESE Q&A IN ONE PLACE? ✨📚



OFFICIAL WEBSITE LAUNCH

- 🌐 ACCESS THE COMPLETE NOTES ON THE JAVA TECH COMMUNITY WEBSITE.

- 📅 **LAUNCH DATE: DECEMBER 22ND, 2024**
DON'T FORGET TO MARK YOUR CALENDAR! 🚀

STAY UPDATED

- 📱 UPDATES WILL BE SHARED IN THIS COMMUNITY CHANNEL.

FOLLOW US ON INSTAGRAM

@JAVA_TECH_COMMUNITY FOR REAL-TIME UPDATES!